

Geometric Query: E0 vs 3D-Position Prediction

(a) E0 input query

Input to E0 route

Query:

"the bench is between two large bookshelves . the bench is rectangular and red brown in color ."

VLM evidence:

- candidate RGB canvases
- visual appearance and local context

(b) 3D-position input query

Input to 3D-position route

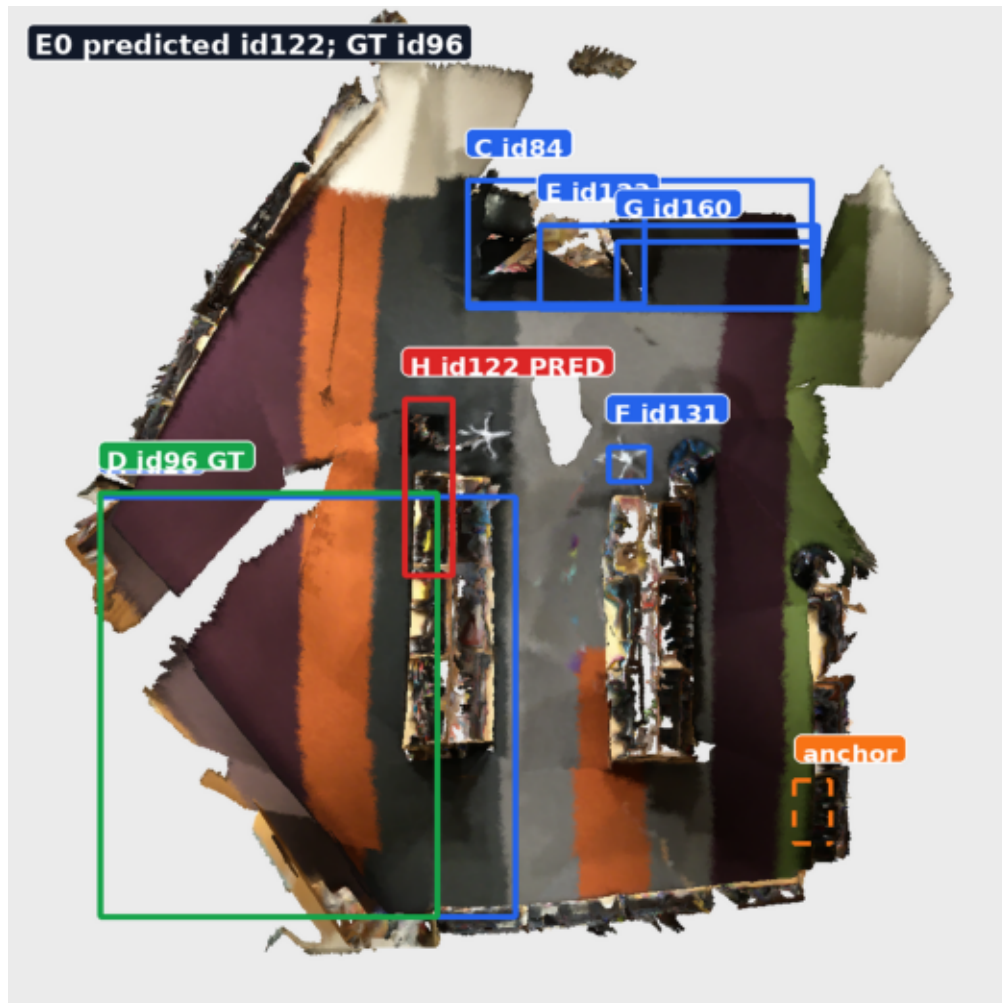
Query:

"the bench is between two large bookshelves . the bench is rectangular and red brown in color ."

Structured evidence:

- candidate-anchor offsets
- distances to the bookshelf anchor
- candidate sizes / heights

E0 prediction on actual mesh BEV render



3D-position prediction on actual mesh BEV render

